2 Information and Communication Technologies Based Technologies and Systems

Information and Communication Technologies (ICT) are classified as traditional and modern ICT. Both of these include the tools, equipment, and support.

Traditional Information and Communication Technology

The means and media which are included in traditional ICT are

- Printed media such as textbooks, resources books, journals, magazines, newspapers, dictionaries, etc.
- Verbal information and ideas are exchanged with parents, teachers, and the members of society.
- Audio–video hardware material such as radio, TV, motion picture, etc.
- Three dimensional aid material such as a model, mock-up, puppetry, etc.
- Graphical aid material such as maps, posters, charts, diagrams, etc.

Modern Information and Communication Technologies

These are not similar to traditional technologies; they include a combination of hardware and software, media, and delivery system. Modern ICT has now become digital. The various components of modern ICT are

- Multimedia personal computer, laptop, and notebook
- Multimedia projector (LCD or DLP) to communicate with a large group
- Multimedia personal computer/laptop with a video card and web camera or digital camera
- Digital video camera
- Digital library
- Computer database and data processing mechanism, CD, ROM, and digital versatile disk (DVD)
- E-media such as FAX, the internet, and World Wide Web (www)
- Hypermedia and hypertext resources
- Local area network, metropolitan area network, and wide area network (LAN, MAN, and WANN)
- Video text, tele text, interactive video text (IVT), interactive video disk (IVD), and interactive remote instruction
- The idea of virtual classrooms and virtual reality
- Application software such as word processing, spreadsheets, PowerPoints, simulation, and speed recognition
- Cyber cafe
- Navigation systems such as global positioning system (GPS) and NAVIC
- Augmented reality and virtual reality
- Mobile
- Television

Computer

It is an electronic device that takes input from its user, stores, retrieves, and processes a large amount of quantitative and qualitative data, and generates the required output as per the processing instructions given to it by the users with high speed, accuracy, and reliability.

Structure of Computer

The primary function of the computer is to process the data. It takes data from input devices, performs the operation, and gives the data to output ports. The main parts of a computer include

- Input unit
- Output unit
- Central processing unit (CPU)

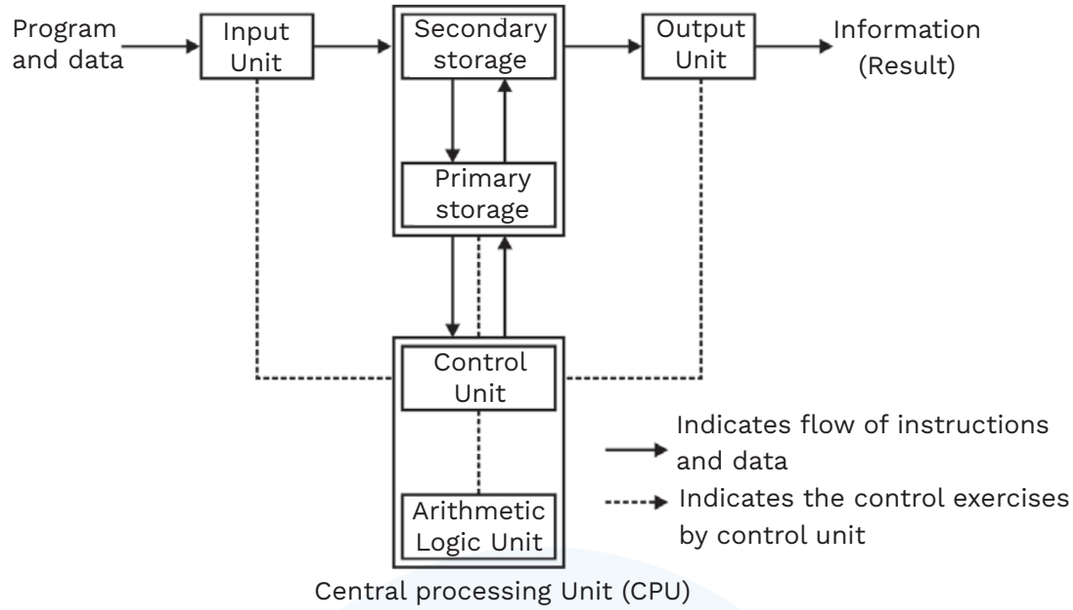


Fig. 2.1

Input Unit

- The first part of a computer is the input unit, which takes in information and commands from the user. These tasks are completed with the use of input devices. There are many different types of input devices, including
- The term 'optical character reader' (OCR) refers to a device that can scan text written in a non-standard typeface on standard printing paper. By shining bright lighting on the inked parts, a logical system may determine which of the many potential characters is being inspected.
- You may think of a scanner as a cross between a photocopier and a fax machine. A xerox duplicates the original page onto paper, whereas a scanner converts the article copy into an online document that may be edited and customised as needed.
- A light pen may serve as both a writing instrument and a pointing tool. To create the light pen, a photocell was placed inside of a pen shaped frame. The pen detects illumination from a constrained area when placed before a screen's graphical component. Moving the light pointer over the display creates lines.
- Among the popular forms of input method is also the Bar Code Reader. An item with a bar code has a series of parallel bars imprinted on it. The information of the bar code is retrieved by reflecting the laser beam of the bar code reader on the bar code. A light sensor picks up those unique bars at the beginning and end of the picture. Following this, the bar sequence is encoded as a numerical representation, which may then be used for further processing in any way desired.
- Human readable characters are written on paper by utilising a special magnetic ink and then read by a device called magnetic ink character reader (MICR). Banks are the primary users of MICRs.
- A digital camera, sometimes known as a digicam, is a camera that saves pictures electronically instead of on film. When light enters a digicam, it is detected by light sensitive sensors, which then convert the data into digital form. The device's memory stores the resulting photos.
- A transmitter known as a microphone allows us to record and playback audio on a system. To record sounds on a system, a microphone must digitise them.

Output Unit

- The output unit is in charge of transforming the data into a form that can be consumed by the end-user. An output device's job is to show the user the results of some kind of processing.

- In the case of the cathode ray tube, functionality is similar to that of a standard tv set. The glass tube's rear end conceals an electron cannon. Using this method, electrons are shot at arrays of phosphor spots that line the interior of the display's screen, and these Ph dots emit their respective colours when hit by electrons, creating a glowing effect.
- When it comes to computer outcomes, the Monitor is the king. It shows data in a manner comparable to that of a tv set. Pixels (represented by pixel value) are the tiny coloured dots that make up an image on a computer screen.
- Sometimes termed liquid crystal display for the material they use, and other times named TFT displays for the tiny transistors that make them function. Every pixel of a liquid crystal display (also known as a 'TFT' display) is made from liquid crystal. Depending on the voltage applied across it, light may be polarised. This capability enables the generation of pixels. Each LC cell is a 'pixel' in this system.
- An output device, or printer, prints data from a computer onto paper. There are two types of printers.
- Physical engagement between the ink cartridge and sheet is used in Impact Printers. The impact printer family has several subsets, including daisy wheel printers, line printers, and dot matrix printers.
- Dot matrix printers are now a widely used kind of impact printer, which produces a single character at a time. There is a pin array in the printing head of a DMP. Selected pins shoot against inked ribbons as the head travels over the paper, creating a pattern of dots. Due to the speed constraints of electrochemical devices and the need to keep printing costs down, a new kind of printer known as a non impact printer has emerged.

Secondary Memory Storage Devices

These types of memory are the additional memory devices used to permanently store the data or programmes. In other words, they are known as magnetic surface recording devices. Some commonly used secondary memory devices are given.

- Floppy disk drive (FDD)
- Hard disk drive (HDD)
- CD ROM's drive (CD)
- Digital versatile disks (DVDs)
- Blue ray disk (BD)
- Pen drive (USB)

Floppy Disk Drive

The floppy disk drive (FDD) was the primary means of adding data to a computer until the CD-ROM drive became popular. FDDs have been a key component of most personal computers for more than 20 years. A FDD reads and writes data to a small, circular piece of metal coated plastic similar to an audio cassette tape.

Hard Disk Drives

- Hard disk drives (HDD) are a common form of data storage on most computers. It stores the operating system software such as word processors, spreadsheets, databases, and data. Data are stored by magnetising the surface of a flat, circular plate. These plates rotate at high speed (60–120 rpm). A read/write head floats on a cushion of air a fraction of a millimetre above the disk's surface. The drive is in a sealed unit to protect it.
- Hard drives usually consist of several disks on a single spindle. Each disk can store data, which has the advantage that each surface can have its own read/write head, as shown in the diagram above. This also enables the read/write heads to operate simultaneously, so data can be transferred more quickly than using one larger disk. A typical PC will have a hard disk capacity between 120 GB and 1 TB.

CD-ROM Drive

CD-ROM drives (CDDs) use the same technology as music CDs and CD players. The data are stored on the disk digitally, and a laser beam is used to read the data on the disc. Because light is used to read the disc, the data may be packed closely and the disc, therefore, has a large capacity. Because of their huge storage capacities, they are useful for distributing software, storing clip art, and use in multimedia systems. CD-ROMS are read-only discs, meaning you can read data

off the disc but not alter it or store new data. A CD recordable (CD-R) drive has a read/write capability. Standard CD-R discs can be written on only once, but there is a type of disc called CD-FW that can be written, erased, and rewritten with a suitable drive.

Digital Versatile Disks

Digital versatile discs (DVDs) are similar to CD-ROMS in which they look identical and are used to store store more data. Similar to CD-ROMs they are used to store computer application software, multimedia programs (encyclopedias), and full length movies. DVD can replace six to seven CD-ROMS. A CD-POM stores 700 MB of data, whereas a DVD typically stores 4.7 GB (4700 MB). DVDs are used to store movies and produce much better quality pictures and sound compared to normal videotape.

Pen Drives

Compared to floppy discs, pen drives are far more portable; the average pen drive weighs less than an ounce and may be removed and written several times (30 g). With continual advancements in size and price per capacity, today's storage options span the gamut from 64 MB to 256 GB. A pen drive combines a Universal Serial Bus (USB) interface with NAND-type flash memory to store information. Some support USB 1.1 or USB 2.0 connections and offer a 10 years data retention period in addition to allowing 1 million write or erase cycles. There may be benefits to using a USB flash drive rather than a floppy disc or another portable storage device. Their absence of moving components allows them to be smaller in size, quicker in processing, able to store more information, and more reliable in operation. There is also a trend toward selling computers without floppy disc drives, as USB ports are standard on almost all popular computers and laptops nowadays.

Blue Ray Discs

The blue lasers used to read a Blu ray Disc (BD) provide a higher data density than the relatively long red lasers used to read DVDs, thus the BD. The Blu-ray Disc (BD) was developed to replace the DVD format as the preeminent optical disc storage medium. Similar in size to CDs and DVDs, the plastic disc measures 120 mm in diameter and 1.2 mm in thickness. Standard BDs have a capacity of 25 GB per layer, with dual layer discs having a capacity of 50 GB being the norm for full length motion pictures. Discs in both triples with 100 GB storage and quadruple with 128 GB storage layer configurations are compatible with BD–XL rewriters.

In addition to the technical requirements, a group of multimedia file formats is often linked with BDs. BDs are mainly used to store and playback video content like full length movies. In most cases, the visual and audio quality can be improved beyond what can be achieved with a DVD using one of these formats.

Memory Hierarchy

- A memory hierarchy is used in a computer, and high-speed memory (cache memory) is used before the primary memory operation.
- Secondary memory is non volatile and has a huge capacity at a low cost.
- The speed of a computer is represented in terms of Floating operations per second (FLOPS), i.e., the number of operations it can perform in a second.
- Computers have no IQ, we have to give input commands to process it, but the speed of operation is very fast.

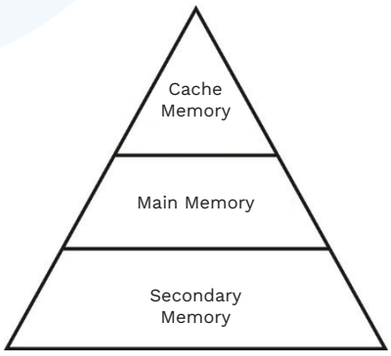


Fig. 2.2

Computer Memory

Memory is that part of a computer that is used store data and information. Computer memory is the storage space in the computer, where data is processed, and instructions required for

processing are stored. The memory is divided into a large number of small parts called cells which has a unique address. The memory system is a hierarchy of storage devices with different capacities, costs, and access times.

CPU Registers

- Internal registers in a CPU are temporary storage areas for data instructions. They are not a part of memory, rather they are special additional storage locations that offer the advantage of speed
- Registers work under the direction of the control unit to accept, hold and transfer data and perform arithmetic or logic operations at a much faster speed. Accessing data internal register is the quickest way to access memory.
- Computer usually assign special roles to certain registers such as
  - Accumulator: a collection of a result of the computation.
  - Address register: keeps track of instructions stored in memory.
  - Storage register: temporarily holds data taken from or about to be sent to memory.
  - General purpose register: for several other functions.

Cache Memory

- The next level to register is the cache memory near the CPU. It acts as a buffer between the CPU and the main memory. It holds those parts of data instructions that are most frequently used by the CPU.
- There are often two or more levels of cache. A cache can be further subclassified as L1, L2, and L3 cache L1 cache is accessed without delay, whereas L2 and L3 take more clock systems to access, respectively.
- Cache is much faster than main memory and consumes less access time. However, cache memory has limited capacity and is very costly.

Primary Memory (Main Memory)

- Primary memory stores the data and information currently being used by the computer. The data gets lost after the power is switched off. It is usually made up of semiconductor devices aid hold limited capacity.
- Primary memory can be further sub classified into two categories RAM and ROM.

Random Access Memory (RAM)

- RAM is the internal memory of the CPU for storing data, programs, and output RAM is a temporary storage that the processor uses to store programs while they are running.
- The idea of a program running from a computer's main memory is known as the stored program concept'. When invoked, a program will be loaded from the hard disk into RAM, and any data entered is stored in RAM.
- RAM is a voltage memory, which means its content is lost as soon as the machine is turned off.
- Access time in RAM is independent of the address, that is, each storage location inside the memory is as easy to reach as other locations and takes the same amount of time.
- Generally, RAM capacity in a typical computer varies from 256 MB to 8 GB installed. The more the installed memory, the faster system will be.
- RAM is of two types: static RAM (SRAM) and Dynamic RAM (DRAM).

Static RAM (SRAM)

- Static RAM can retain contents in its memory by having a constant source of power applied. However, data is lost when the power goes off due to its volatile nature.
- SRAM consists of a matrix of 6 transistors and no capacitors. As transistors do not require power to prevent leakage, SRAM need not be refreshed on a regular basic.
- The application of SRAM is more likely to be on the processor itself or m the form of small but high-speed cache memory.

Dynamic RAM (DRAM)

- DRAM is the most common type of main memory found in computer systems. This type of memory is dynamic because its storage coils must be refreshed. This is done by placing the memory on a refresh circuit that rewrites the data several hundred times per second.

- DRAM can store more data and is, therefore, more suited to low-cost, high-capacity systems. All DRAMs are made up of memory cells, which are composed of one capacitor and one transistor.
- Synchronous dynamic random access memory (SDRAM) is a special type of DRAM. It is faster than asynchronous memory because the memory is synchronised with the system bus and clock and, therefore, the processor.

SRAM	DRAM
Long life	Short data lifetime
Periodic refresh is not necessary	Need to be refreshed continuously
Faster than DRAM	Slower than SRAM
Large size and high cost	Small size and less expensive
Used as cache memory	Used as RAM
High power consumption	Less power consumption

Table 2.2 Difference between SRAM and DRAM

Read Only Memory (ROM)

- ROM is a special type of non volatile memory permanently built into the computer during production.
- The information from ROM can only be read, and it is not possible to write new information into it. It permanently stores a set of instructions which instruct the computer on how to work. This operation is referred to as bootstrap.
- After the computer is switched on, it uses instructions stored to carry out a series of tasks automatically before the computer is ready for use.
- ROM can be further categorised as
  - Programmable read-only memory (PROM)
  - Erasable programmable read-only memory (EPROM)
  - Electrically erasable programmable read-only memory (EEPROM)

Programmable Read-Only Memory (PROM)

- PROM is a read-only memory that can be modified by a user only once.
- This type of ROM can be programmed using a special device called a PROM programmer. Once their data is programmed, it becomes permanent and cannot be changed.

Erasable Programmable Read-Only Memory (EPROM)

- EPROM is a type of ROM that can have its contents erased using ultraviolet light. It can be then reprogrammed by an RPROM programmer.
- During programming, an electric charge is trapped in an insulated gate region, which can be erased by exposing the change to ultraviolet light.
- The reprogramming procedure can be carried out multiple times; however, constant erasing and rewriting will eventually render the chip useless.

Electrically Erasable Programmable Read-Only Memory (EEPROM)

- EEPROM is programmed and erased electrically, and the process can be continued about ten thousand times.
- This type of RUM works similarly to flash memory in that its contents can be ‘fashed” for erasure and then written to without having to remove the chip from its environment.
- EEPROMs are used to store a computer system BIOS and can be updated without returning the unit to the factory.

Motherboard

- The motherboard is a part of a computer which serves as a single platform to connect all the parts of a computer. It is considered the backbone of a computer.
- The motherboard connects the CPU, memory, hard drives, optical drives, video card, sound cards, and other ports and expansion cards directly or via cables.

- Motherboard contains ports to connect all of the internal components. It provides a single socket for the CPU, whereas for memory, normally, one or more slots are available.
- Some ports are available to attach the hard drive and optical drives. Motherboard also carries fans and a special port designed for power supply.
- Memory unit: the memory unit is the amount of data that can be stored in a storage unit. This storage capacity is expressed in terms of bytes

Bit = Binary Digit (0 and 1)	1 Megabyte (MB) = 1024 KB
1 Nibble = 4 bits	1 Gigabyte (GB) = 1024 MB
1 Byte = 8 bits	1 Terabyte (TB) = 1024 G8
1 Kilobyte (KB) = 1024 bytes	1 Petabyte (PB) = 1024 TB

Arithmetic and Logic Unit (ALU)

- It performs all arithmetic operations (addition, subtraction, multiplication, etc.) and all logical operations. (greater than, less than, equal to, etc.).
- It works on the basis of the control signal as given by the control unit and intermediate results of the processing are stored in the temporary registers of ALU.

Control Unit

- This unit plays a vital role in any computer system as it is used to produce the control signals to manage and control all the activities of all other units. It acts as a central nervous system for a computer.
- This unit receives instructions from the main memory, interprets, and executes the instruction, and transfers the data from the instructions to ALU for processing.

Concept and Types of Software of Computer

- Software refers to something intangible. Although often of great value, the software cannot be picked up or weighed. However, software is required to make hardware do anything. Likewise, software needs hardware to be of any value. 'Typical examples of the software include the web browser you use to read a page, the web hosting software used to show you a page, and the operating system (Windows, Mac) that you interact with whenever you use your computer.

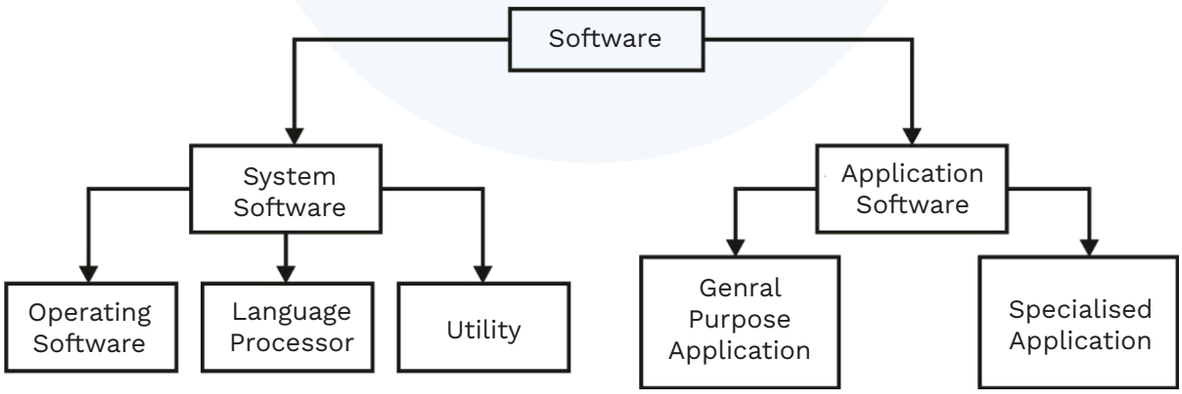


Fig. 2.3

- Software is the collection of programs that make the hardware do its job. Software is usually divided into two major categories
  - System software
  - Application software

Concept of System Software

- What we mean by 'system software' is the software that makes it possible for the various components of the system to interact with one another. The system cannot function as

a whole because of the lack of a software system. Data transfers between memory and disc and text rendering onto a monitor screen are examples of the kinds of activities performed by system software.

- Computer resource software solution at the lowest level, including the OS and any utility applications (such as Compiler, Linker, etc.) In this context, 'system software' refers to the software that runs on computers. Well-known examples of system software include operating systems like Mac OS X and its versions, Windows, and Linux.
- Systems software's goal is to shield application developers from having to know specifics about the computer's memory, processor, and other hardware capabilities, as well as the specifics of any communications or any hardware devices such as displays or printers, etc.
- A software suite is often the bare bones of a computer, and may not be a main reason a consumer would purchase a device. System software is further required that helps a consumer operate the system. Software such as language translation, data communication software, data management programme, and a compiler are all part of this category.

Types of Software

Shareware

- Shareware software is copyrighted software distributed on a try before we buy basis by paying a nominal shareware fee.
- Users who want to continue using the program after the trial period are encouraged to send a payment for the program to the program developer.
- Shareware is generally less expensive because often, a single programmer develops it.

Freeware

- Freeware programs are programs that are given away free of charge and often made available on the Internet.
- The programmer offers programs as freeware either for personal satisfaction or to assess its reception among interested users.
- Program developers often retain all rights to their freeware, and users are not free to copy or distribute or sell it further.

Open Source Software

- Open source describes the principles and methodologies to promote open access to the production and design process for various goods, products, resources, and technical conclusions or advice.
- The term is most commonly applied to the source code of software that is made available to the general public with either relaxed or non-existent intellectual property restrictions.
- This allows users to create user generated software content through either incremental individual effort or collaboration.

Performance of Computer

The performance of computers is measured in terms of the following parameters.

CPU Speed

- The CPU contains a quartz clock. Each time this clock 'ticks', one instruction can be dealt with by the CPU. So the more times this clock ticks per second, the more instructions the CPU can carry out and the faster things get done.
- The speed of the CPU is measured in either Megahertz (MHz) or, more commonly now, in Gigahertz (GHz). A 1 MHz CPU can carry out one Million Instructions Per Second (MIPS). A 1 GHz CPU can carry out 1 billion instructions per second. A typical CPU installed in a computer today would run at around 3 GHz, meaning it can carry out 3 billion instructions per second.
- Therefore, we can computer performance might be to upgrade your CPU to a faster one, especially if you have a reasonably slow, old machine.

Memory Size

- The CPU uses a temporary memory (called RAM) that stores documents and files while

you are working on them. Therefore, if you have a word document open and an email, a website, and a couple of images, they are all sitting in RAM. You can be sure of this because remember, when you switch your computer off, it will disappear RAM is volatile, i.e., non permanent memory.

- This can cause a problem over time because the RAM in your computer will eventually fill up, and your computer will appear to slow down.
- If you want to improve your computer's performance and only have a small amount of RAM, such as 512 Mb or even 1 Gb, you could add extra RAM chips.

Number of Applications

- The more applications, files, and documents you have open, the faster your RAM will fill up. Even if you increase the amount of RAM you have in your computer, it will eventually fill up during the day.
- Therefore, another way of improving the performance of your computer is to limit the number of applications you have running at any one time and the number of files/ documents you have open. As soon as you have finished using something. Close it down and release some of the RAM elsewhere.

Types of Computer

Here in this section, we have classified computers on the basis of size and storage, but there is another classification of computers.

Programmable Computers

- The power, memory, and size of programmable computers vary widely. They can link to global computer networks to share information independent of locale and serve as notebooks, schedulers, and mailings.
- Personal Digital Assistants abbreviated as PDAs, the tiniest of these devices, can even be handled with one hand. Salespeople, educators, etc., are the most common users.

Workstation

Like PCs, but with additional storage and computation power, workstations provide a variety of advantages for professionals. Coupled with other work terminals or desktop computers, they are used in research, corporate and industrial settings that need elevated levels of processing capability.

Embedded Computers

There is a wide variety of system sizes and processing speeds available, which brings us to the concept of embedded computers. These are often configured to do a single, discrete job, such as tuning to a certain television channel or maintaining precise time. The tiniest ones may be found deep inside the circuits of everyday items like TVs, washers and dryers, autos, and clocks.

Laptops and Personal Computers

- It is possible stow away these notebook looking laptops and other portable computers of these categories in a briefcase. Laptops get their name from being used on a person's lap, whereas desktops get theirs from being used on a desk.
- Desktop PCs and laptops are prevalent both in workplace and home where they are used for everything from word processing and financial tracking to playing games and chatting with coworkers.
- Laptops are portable PCs comparable to personal computers in terms of hardware and software but distinguished by their smaller size and fewer weight thanks to the use of flat, lightweight liquid crystal display screens. Along with a keyboard and a scroll pad for inputting data.

Supercomputers

When predicting weather forecast, for example, sophisticated and time consuming calculations need to be performed; this is where supercomputers come in. These computers partition jobs into smaller chunks, which are then processed by individual CPUs to boost throughput and performance. Parallel processors are the technical term for such machines. Companies of all sizes, as well as academic and research organisations, and the armed forces employ them.

Mainframes

Mainframes are utilised in research and to manage enterprises and factories. Services such as utilities, banks, and other financial institutions rely heavily on them. Mainframes, unlike workstations, are designed to serve several users simultaneously over a network of other PCs.

Mobile Network Technologies

Global System for Mobile Communications

- An electronic system is based on time division multiple access (TDMA) technology, the global system for mobile communications (GSM) is a global platform for wireless phone communication.
- TDMA divides a small spectrum of 30 kHz in width into three separate time slots, each of which is 6.7 ms long. One third of the radio time is allocated to each discussion. This is made feasible by the compression of transformed speech data to electronic information, which allows for far more efficient transmission. This means that TDMA can handle three times as much data as an analogue system for the same multitude of channels.
- There are several advantages to using GSM systems. Data networking, telecommunications encryption, multi-party conferencing, call forwarding and waiting, caller ID, and SMS services are some of them.

Code Division Multiple Access

- CDMA is an umbrella term for a family of technologies used in 2G and 3G wireless networks. As the name suggests, code division multiple access (CDMA) is a kind of multiplexing that enables many signals to share a single transmission channel.
- Cellular phone networks operating in the 800 MHz and 1.9 GHz ultra-high frequency bands use this technology. CDMA uses distributed spectrum technique in conjunction with an analogue to digital conversion.
- The initial step in processing audio is to convert it to binary data. Then in the next step, the frequency of the broadcast signal is modulated in accordance with a predetermined pattern or code, making it accessible to only those receivers whose frequency response has been similarly programmed. It is tough to clone someone with a frequency sequencing code since so many of them exist. Due to CDMA's interoperability with other cellular standards, users may freely move between networks throughout the country.

CDMA vs GSM

- The main differences between these technologies are worldwide interoperability and how the networks manage the activation of connections.
- Carrier activation of CDMA phones is performed over the air using the ESN (equipment serial number). Most CDMA carriers can reject activation of phones not meant for their network due to the existence of databases listing all ESNs permitted for usage on each company's network.
- However, the activation process for GSM phones is different. There is a Subscriber Identity Module (SIM) card connected to each user's account. The encrypted version of all the information necessary to disclose your wireless account to the network is stored on this card. Not only can you use a wider variety of phones with your service, but it also allows you to take any GSM hardware and make it 'your phone' instantly, including the ability to carry your contacts right on the SIM card.
- You could literally take your phone, smash it with a hammer, extract the SIM card, and with a backup phone in your glove compartment, be back up and running in 30 s. Or you could carry a pre-paid SIM with you and pop it into your phone in case there was ever a time when you wanted to have phone service but not be reachable by your usual number. GSM's strength is in its flexibility. That's not to say CDMA providers do not have advantages. In the United States, at least, Sprint and Verizon are well ahead of AT&T and T-Mobile on deploying high-speed Internet. Both CDMA carriers have reasonably extensive wireless broadband

footprints, while AT&T is relatively small, and T-Mobile is still preparing to launch such a system. Thus, some users in the United States have to choose whether flexibility now is worth waiting for the broadband coverage to catch up.

Mobile Phone Generations

Mobile Radio Telephone (or OG)

- It was mobile radio telephone systems rather than cellular mobile telephony that came first. In 1946, Motorola and the Bell System launched the world's first commercial cellphone service, known as Mobile telephone system (MTS), as a branch of the wired telephone business.
- These systems are often retrospectively referred to as pre-cellular or zero-generation systems since they preceded the first development of cellular cellphones. Although briefcase versions were available, most of these mobile phones were installed in automobiles and trucks.

1G

- The first version of mobile telecommunications technology is often referred to as '1G.' This refers to the analogue generation of mobile networks that emerged in the 1980s and persisted until the advent of 2G digital networks.
- The primary distinction between the 1G and 2G mobile phone systems is that the 1G networks utilise analogue radio signals and the 2G networks employ digital ones. However, the radio towers in both systems are linked to the rest of the telephone network by digital signaling.

2G

- Second-generation mobile telecommunication technology is abbreviated as '2G.' Finland was one of the first countries to introduce GSM based 2G cellular telephony networks commercially.
- First, 2G networks were more secure than their predecessors because phone communications were encrypted using digital technology. Two generation (2G) systems were substantially more spectrum efficient, providing for extremely high rates of cell phone adoption. The short message service (SMS) was the first data service for mobile phones made possible by the second generation of mobile networks.

GPRS

- To bridge the gap between 2G and 3G in mobile wireless communication, 2.5G (also known as GPRS) was developed. The term '2.5G' (second and a half generation) refers to 2G systems that also employ a packet switched domain with the traditional circuit switched one.
- Data transmission through classical circuit switching is invoiced per minute of connection time, regardless of whether the user is actively using the capacity or is in an idle state, while GPRS data transfer is often charged per megabyte of traffic sent.
- In addition to using part of the already in place 2G infrastructure in GSM and CDMA networks, 2.5G also offers some of the features of 3G, such as being packet switched. General Packet Radio Service (GPRS) is the most well known example of a 2.5G technology. Services such as email and Web browsing, as well as more advanced MMS, are all possible with GPRS's Internet connectivity.

EDGE

- A 2.75G (EDGE) cellphone is a circuitry shifted digital mobile phone, unlike a 2G phone. For systems that are billed as 3G but did not match the specifications (such as CDMA 2000 without multi carrier), the name 2.75G has been settled upon. For example, EDGE based networks.
- Improved data transmission rates are made possible by the addition of an enhancement to regular GSM called enhanced data rates for GSM evolution (EDGE), enhanced general packet radio service (EGPRS), or IMT single carrier (IMT-SC).

3G

- Third-generation mobile telecoms (3G) is a set of guidelines for smartphones and telephone services that meet the requirements of the International Mobile Telecommunications-2000 (IMT-2000) set by the International Telecommunication Union (ITU).

- In order to support the high data rates required by 3G networks, mobile carriers had to invest in entirely new infrastructure and get licences for whole new radio frequencies. Costs associated with updating broadcasting gear contributed to further setbacks, notably for UMTS, whose rollout necessitated the renovation of the vast majority of rooftop antennas. Many providers could not or postpone their purchase of these modern technologies because of the aforementioned problems and challenges with installation. It took a while for 3G to gain traction throughout the world.
- The state owned MTNL launched 3G enabled Cellular and Broadband services in Delhi and Mumbai in December 2008. BSNL, another government operator, introduced 3G operations in February 2009 in Chennai, Tamil Nadu, and subsequently expanded them to the rest of the country. It was made public in April 2010 that 3G wireless frequency would be auctioned off, and in September of the same year, all commercial carriers were given access to this spectrum. Cell phone users now had access to new types of software made possible by the increased capacity and geographical data provided by 3G handsets.

3.53G

- High-speed downlink packet access (HSDPA) is a mobile phone protocol that represents the third generation (3G) of wireless technology.
- HSDPA is a packet-based data service in the W-CDMA downlink that can transmit up to 8–10 megabits (and 20 megabits/s for MIMO systems) across a 5MHz bandwidth.

4G

- Known as the fourth generation of mobile communications standards, or simply '4G,' this technology is rapidly becoming prevalent. A 4G network may connect mobile devices such as laptops, cellphones, and tablets to the Internet at ultra-high speeds. Potential uses include high-speed mobile internet connectivity, Voice over internet protocol, gaming services, mobile TV in high definition, video conferencing, and 3D television.
- The International Mobile Telecommunications Advanced (IMT-Advanced) specification, developed in March 2008 by the International Telecommunications Union-Radio communications sector (TTU-R), established peak speed requirements for 4G service at 100 Megabits/s for high mobility communication such as from trains and cars and 1 Gigabit/s for low mobility communication such as pedestrians and stationary users.
- In contrast to 3G's dual mode foundation of circuit switches and packet switches at various network nodes, 4G will rely only on packet switching. Compared to previous versions, 4G networks do not enable circuit switched telephones but rather IP based telecommunication, including IP telephony.
- IPv4 address depletion was predicted to be nearing its end by the time 4G got rolled out. For this reason, IPv6 must be supported in the 4G environment if a massive proportion of wirelessly equipped devices are to be supported. IPv6 eliminates the necessity for network address transition (NAT) by expanding the available pool of IP addresses.
- To connect with devices on current IPv4 networks, NAT will still be necessary. However, this address splitting technique allows a greater number of devices to share a smaller pool of available addresses.
- As the next generation of wireless broadband internet technology, long term evolution (LTE) will fuel the advent of 4G Mobile communications optimised for ultra fast data transfer. This method employs the utilisation of radio frequencies to convey information. Similar to 2G and 3G, LTE can accommodate growing data needs without sacrificing connectivity. For this reason, transitioning to LTE for a 3G provider is straightforward. Compared to 3G technology, LTE based 4G technology data transfer rates are roughly four times as fast.

Multiplexing

- Multiplexing, or mixing, is a technique used in the telecommunications and computer

networking industries to transmit numerous independent analogue message signals or digital information streams across a single, common transmission line. The objective is to pool funding for a distributed system. For instance, multiple calls may be sent over the same line in the wireless industry. Multiplexing may be accomplished using a system known as a MUX or a multiplexer, whereas demultiplexing requires a different system known as a DEMUX or demultiplexer.

- IMUX or Inverse multiplexing, on the other hand, rather than sending a single data stream across several channels, sends numerous streams of data at once. It is basically the inverse of multiplexing.

Types of Multiplexing

Time division multiplexing (TDM), code division multiplexing (CDM), Space division multiplexing (SDM), and frequency division multiplexing (FDM) are various forms of multiplexing technologies that may be distinguished by their own unique characteristics.

Code Division Multiplexing

In CDM or a spread spectrum, the spectral bandwidth is much larger than the bit rate, allowing numerous channels to simultaneously use the same frequency band. Code Division Multiple Access is a channel access strategy that uses Code Division Multiplex technologies to distribute intercell disturbance over a large number of customers. This makes it suitable for usage in cellular phones and wireless networks. Along with this, another common application is the GPS which uses CDMA.

Frequency Division Multiplexing

The technology behind FDM is fundamentally analogue. By transmitting signals in several different frequency ranges across a single channel, FDM can combine numerous digital signals into a single signal. Broadcasting television through cable is a typical use case for frequency division multiplexing (FDM). Although there is only one cable running to a client's house, many tv stations may be sent down one connection simultaneously. To get the intended broadcast, recipients must adjust to the correct frequency.

Space Division Multiplexing

- SDM essentially means using separate point to point lines for each channel. A phased antenna array is a collection of radio components used to perform SDM in wireless communication. Multiple input, multiple output, known as MISO multiplexing, is one such example.
- An analogue stereo audio wire is one such instance. Using a multi pair telephone line and separate cables for the left and right channels. The analogue telephone access network is a different type of switching star network.

Time Division Multiplexing

TDM is a digitised and, in certain situations, analogue technology. For TDM, it is necessary to sequentially order sets of several bits or bytes out of each input stream so that they may be mapped to the correct receiver.

Television

- Cathode ray tube technology or CRT technology-dominated tv sets for a long time. LCD, plasma, LED, and OLED TVs are some of the recent technological advancements that have occurred in this space.
- In order to show images, LCD, LED, and plasma TVs must upscale them and remove the interlacing, and rather than scanning them straight onto the display like a CRT TV, they retain the images and insert them onto the screen one frame at a time. Specific characteristics of each category of these devices are detailed below.

Liquid Crystal Display TV

- The development of LCD technology finally allowed television sets to ditch bulky and outdated CRT displays. Cold Cathode Florescent Lamps (CCFLs) and molecules of liquid crystals are the two main components of LCD technology's display. White light from the

CCFLs is utilised to illuminate the molecules of the liquid, and a picture is formed as the light travels through. As a result of its unique method of display, an LCD television offers substantial advantages over conventional televisions

- Improved clarity
- Improvements in the viewing angle, up to 175 degrees
- Improved clarity
- Insensitive to 'burn-in' because of the lack of movement in the pictures
- A decrease in electricity usage
- Excellent for spaces that already have plenty of ambient light

- Negative aspects of this technique include a lack of contrast and difficulty displaying an authentic black image.

Plasma TV

- With this method, neon and xenon gas atoms are used to form a picture. In response to the limitations of liquid crystal display (LCD) televisions, the plasma TV was developed. As a result of its sophisticated colour creation function, which can produce genuine black, plasma TVs provide a wide palette of colours.
- Compared to an LCD TV, the sight angles are much larger, going as high as 180 degrees. Plasma televisions also last longer than their LCD counterparts.
- The technology's drawbacks include a reduced viewing experience in dark situations, a decreased maximum resolution, and an unfixable backlight.
- The viewing distance and angles are also wider, ranging from 160 and up to 180 degrees, which is more than what can be achieved with an LCD TV. A plasma TV also has a higher lifespan than LCD TV. The downside of this technology is that the viewing capability, especially the brightness, is greatly diminished in poor lighting conditions, lower resolution, and irreparable backlight.

Light Emitting Diode TV

LED TVs to represent a technological leap forward in the history of television sets. This is essentially a new generation of LCD TVs, with backlighting provided by LEDs rather than the more commonplace CCFLs or the Cold Cathode Florescent Lamps. An LED television's benefits include

- It has a much reduced energy footprint.
- Excellent image quality that can be seen from a variety of angles is a hallmark of an LED TV, which is made possible by its superior colour reproduction and contrast ratio.
- It remedied a common issue brought on by fluorescent lighting, which was the appearance of multiple dark areas.
- LED televisions are superior to both liquid crystal displays and plasma televisions in every way except price.
- It can withstand shock better than its predecessors and lasts far longer.

Organic Light Emitting Diode TV

- OLEDs are being utilised to make digital screens for a wide variety of electronics, including televisions, computers, mobile phones, and gaming consoles. The creation of white OLEDs for use in solid-state lighting applications is a hot topic of study.
- An OLED is a kind of LED in which an organic compound film serves as the emissive electroluminescent layer and produces light when an electric current is passed through it. This organic semiconductor layer is positioned between two electrodes. One of these electrodes, often both, is see-through.
- It is possible to classify OLEDs into two broad categories
  - Those that use polymers.
  - Those that are based on small molecules.
- It is possible to use either a passive matrix, i.e., PMOLED or an active matrix, i.e., AMOLED addressing method in an OLED display. A light emitting electrochemical cell (LEC) with a somewhat distinct mode of operation is created when moving ions are added to an OLED.
- The active matrix OLED (AMOLED) allows for better resolution, quality, and bigger display dimensions at the expense of requiring a slender transistors backplane to turn on and off each pixel individually.

- A backlight is not required for an OLED screen. Because of this, it is possible to achieve deeper black depths and smaller and lighter dimensions than with a liquid crystal display (LCD). In low ambient light settings, the contrast ratio of an OLED panel will be greater than that of any LCD, including those with cold cathode fluorescent bulbs or LED backlight.

Advancements in Television

Direct to Home Technologies for Broadcasting

- For a long time, direct to home (DTH) satellite television has been all the rage in the media business. Until recently, India only had access to conventional, cable, and satellite (C band) broadcasting options. Broadcasters are on the hunt for an economically and technologically feasible means in light of the increasing preference from consumers for specialised entertainment and the increasing need to preserve rapidly shrinking frequency bands.
- This is a two-step procedure. As a first option, there is 'uplinking', in which a broadcaster sends electronic information through signals representing television and radio shows to a specific transponder within a satellite. The transponder boosts the information and modulates it so that another device may receive it.
- The next phase involves an 18 inch dish antenna picking up signals broadcast by transponders and sending them to a set-top box, which then reformats the signal for display on a television. This is known as 'downlinking' this device, which serves as both a receiver and decoder, is vital to the operation of a DTH setup.
- DTH enables the broadcast of multiple broadcast platforms using a singular, highly efficient Ku band transponder on a satellite operating at frequencies between 1 GHz and 14 GHz.
- It will help broadcasters since it eliminates the possibility of subscription based business distortion at the hands of middlemen such as cable companies. It's the most reliable method for reaching viewers and collecting payments. Therefore many TV networks are using it.
- Improved broadcast clarity and the value added features offered by this service are what will draw in viewers.
- Thirdly, the station might provide internet connectivity, electronic mail, and video on demand (VOD) services. Using highly efficient transmitters may also be utilised to spread important awareness campaigns. More people will end up buying as a consequence.
- Because it allows for several programs to be broadcast from a single transmitter, it is ideal for usage with tiny dish antennas that may be put in houses.
- Fifth, free-to-air direct to home broadcasting has evolved as a feasible option for public broadcasters like Prasar Bharati to provide tv to every individual in the country.

High Definition Television

- HDTV differs from standard definition TV in the following ways
  - Higher visual resolution
  - 16:9 widescreen as the default
  - The ability to accommodate multi channel audio formats like Dolby Digital
- The higher resolution is the key trait of high definition television. The intermittent scanner of normal tv systems divides each frame into two fields, one for the odd numbered lines and the other for the even numbered lines. Each frame is shown in succession, and the human mind fills in the gaps to form a coherent whole, resulting in the degradation of the visual quality.
- As opposed to being divided into individual fields, each frame is shown in its entirety in a progressive scan. The quality of both systems much exceeds that of regular TV broadcasts. Broadcasting HDTV may be received in either the 720p or 1080L formats. The numbers indicate the vertical image resolution in lines, whereas letters denote whether the signal is a progressive scan (denoted by 'p') or interlaced (denoted by 'T').
- In order to avoid having wider images be improperly broadcast as letterboxed

or panned, HDTV adopts a broad screen resolution of 16:9. Digital picture and audio systems are used in HDTV, making the video and audio clearer than on standard television. The only way to get HDTV is to either buy a TV that has an integrated HDTV tuner or buy a separate HDTV reception that can pick up HD channels over the air.

P2PTV

- The name 'P2PTV' is used to describe a category of peer-to-peer (P2P) software programs whose primary function is to redistribute live video streams over a P2P network.
- These applications are appealing because they may theoretically make any TV station accessible throughout the world by having any user feed the stream into the network, with each user acting as a relay for other users watching the video. These live video streams are often broadcast on TV channels from across the globe, but they may originate from other places as well.
- It is common for the incoming streams to lag behind the primary sources by a few minutes in terms of timing. With P2PTV, users share the bandwidth by transferring video streams in tandem. In most cases, a higher number of viewers means greater visual quality for the stations they are tuning into. Several P2PTV connections may be conceptualised as real time variants of the Bit Torrent design.
- When a user requests a specific channel, the P2PTV client software first contacts the channel's 'tracking server' to gather the IPs of peers that broadcast the channel, then makes contact with this peer group to obtain the feed. The tracker logs the person's address and makes it available to anyone who wants to watch the same channel.

IPTV

- Utilising the Internet protocol suite across a packet switched network such as the internet, internet protocol television (IPTV) delivers television services with the expected quality service and interface, security, and dependability.
- There are three broad categories into which IPTV services may be placed
  - 'Time shifted television' includes 'catch up TV' that rebroadcasts a TV program that originally aired some time ago (usually hours or days), 'start over TV,' and 'rewind TV' that replays the live TV program.
  - Real time broadcasts of television shows, with or without viewer participation
  - The third option is VOD, or video on demand, where users may choose from a library of media unrelated to traditional TV shows.
- Unlike Internet television, IPTV is currently being standardised and is best used via a subscription service that provides data at high routes into end user premises through set-top devices. Since the introduction of online video gathering sites like YouTube, HYBRID IPTV has been more popular for two main reasons.
- The term 'hybrid IPTV' describes a system that combines conventional broadcast TV with video provided through either controlled Internet protocols or an open connection. It's a growing movement in the pay TV and consumer sectors. There has been increasing pressure on existing pay TV providers to let their consumers stream Web media directly to their TVs.
- Since operators' bandwidth is a scarce resource, they have sought creative methods to provide these new services to customers without expanding their networks. Specialist IP based operators, typically telecommunications carriers, have sought methods to deliver digital and analogue services to their businesses without increasing the expense of transmission infrastructure.
- Given these tendencies, manufacturers have created hybrid IPTV set-top devices that combine a conventional broadcast tuner with an Internet connection (often over Ethernet).
- IPTV, an alternative to the traditional Headend in the Sky cable TV package, is one such method. Here, the IPTV provider or ISP receives several TV channels via satellite and

then distributes them to individual customers through IP encapsulation. This may alleviate strain on the POP's Internet trunking while providing customers with a wide range of channel options.

- As a result, TV customers may watch more programs from more providers without buying multiple set-top boxes. With a dual set-top, you may stream television from the Internet in addition to your local broadcast and cable channels.
- Video-on-demand (VOD) and catch-up television (CTV), as well as Web based applications, including video conferencing, gaming, and commerce, are made available to consumers using this technology.
- More material and features may be distributed across a switched IP network. When a user tunes into a TV station or channels via a satellite provider, the material is continuously sent downstream to their set-top box, where the user may make any desired changes.
- As many channels as the customer's chosen telecommunications, cable, or satellite provider can fit into the customer's home's incoming 'pipe.' The operation of an IP network is different from one that uses switches. Stuff is stored centrally and just the content the user chooses is sent to their house.
- Video-on-demand (VOD) is being brought to the television using IPTV technology, allowing viewers to peruse an online program or film repertoire, see previews, and then choose a recording. Once a consumer clicks on a link, playback of the selected content begins almost immediately on their TV or computer.
- The ability to combine and streamline services is another perk of an IP-based network. Through the use of IMS-based solutions, this possibility is enhanced. To produce new, more valuable services, 'converged' ones need the seamless integration of many current services. For example, consider the convenience of on-screen Caller ID, which entails both the availability of Caller ID on a TV and the ability to interact with it (send it to voice mail, etc.).
- With the support of IP based services, it will be possible to combine services and content to provide customers access to it on their TVs, computers, and mobile devices at any time and from any location. Because of IPTV, there is no longer a requirement for a separate infrastructure to be maintained in commercial and institutional settings to provide on-demand and live video services.
- If the streaming data is unstable, IPTV will experience problems like packet loss and delays.
- IPTV needs a minimum bandwidth that can only be provided via a dedicated connection.
- It has the optimal amount of fps (frames per second) for transmitting animation. Because of this, the quality of service provided to a big IPTV user base may suffer due to the constraints of the available connection speed and capacity.
- Packet timing and sequence are crucial to the smooth operation of an IPTV feed. Fortunately, we are now beginning to see solutions to the issue become available thanks to advancements in wireless technology.
- Most IPTV providers currently employ wired house communications technology like Ethernet and CAT5e/6/6a/7/7a/8, rather than wireless ones like 802.11b/g/n, due to the constraints of wireless communication.

Voice Over IP

- Voice over Internet Protocol (VoIP) refers to a set of protocols, methods, and transmission mechanisms for delivering voice and other multimedia sessions across IP networks like the Internet. Commonly used terminology for VoIP includes broadband telephony, internet telephony, broadband phone, and voice-over broadband (VoBB).
- IP telephony refers to digital telephony systems that use IP protocols for voice communication, while VoIP is a subset of IP telephony. IP telephony relies on voice-over-internet protocol, or VoIP, to carry customer calls. Users of mobile devices other than phones may make calls and send SMS messages through 3G or Wi-Fi thanks to the widespread availability of VoIP software on smartphones and other Internet connected devices.

- The term 'Internet telephony' is used to describe the transmission of voice calls, faxes, text messages, and VOIP applications via the Internet rather than the traditional public switched telephone network (PSTN). VoIP calls begin with the signalling and media channel setup, followed by the digitalisation of the analogue voice signal, encoding, packetisation, and transmission of those IP packets via a packet switched network. When the voice stream is received, it goes through a series of operations that are identical to those performed during transmission but in reverse order.
- Skype is a well known proprietary version of a protocol that operates on some of the same principles as P2P networks. Using their already established broadband Internet connections, VoIP service users may make and receive calls very similar to that of the traditional PSTN (public switched telephone network). This trend began in 2004 and has since exploded in popularity (PSTN). VolP phone businesses that provide a full suite of services, including Direct Inbound Dialling for both incoming and outgoing calls, are considered full service. In order to make a call over a VoIP service, you'll need a VoIP phone.
- You may use a phone number that is normally connected with a certain geographic location anywhere in the globe when you use VolP. It also has other sophisticated functions that regular telephone networks do not provide. However, VolP's lower price tag is its most compelling selling point compared to traditional phone networks.
- Moreover, VoIP calls to other countries are often relatively cheap. In addition, this benefit will grow in significance as VoIP use increases, and calls between VoIP users are often free. True VoIP allows its users to make free phone calls to other users.

Satellite Orbits

In passive satellites, the transmitter of the ground station requires enormous power, and the receiving station receives a fraction of that power. Then it is amplified by active electronic means. 'ECHO-I' was the world's best known passive communication satellite launched in 1960.

Satellite Orbit

- The path in which the satellite goes around the earth is called the orbit path, and the distance from the centre of the earth to the satellite is called the orbit radius. An orbit is characterized by two attributes – altitude and inclination.
- Satellites can be classified in terms of altitude.
  - Low earth orbit (LEO) satellite
  - Medium earth orbit (MEO) satellite
  - Geostationary earth orbit (GEO) satellite
  - Highly eccentric orbit (HEO) satellite

Types of Satellite	LEO	MEO	GEO	HEO
Altitude relation period time in sight example uses	500–1500 km 90 min 15 min iridium mobile communication and for surveying	5000–10000 km 5–12 h 2–4 h GPS global communication on such as email, FAX, and telephony	36000 km 24 h always VSAT global communication on such as TV and radio transmission data transmission	15000–30000 km less than 24 h 8 h Molnya communication amount polar countries

Table 3.1 Performance Characteristics of Different Altitude Satellites

Satellite Applications

- There are numerous satellite applications, but they can be broadly classified into the following category.
- Communication: TV, telephony, data transfer such as email and internet, etc.
- Remote Sensing and Earth Observation: can be done with the help of the Lower Earth Orbit (LEO) satellite.
- Meteorological: includes weather surveys to study different layers and amounts of oxygen content in the atmosphere.
- Military: includes short distance local communication from one camp to another, to study the location of the enemy, etc.

Satellite Bands Spectrum

Band	User	Downlink Bands (in GHz)	Uplink Bands (In GHz)
UHF	Military	0.25–0.27	0.29–0.31
C band	Commercial	3.7–4.2	5.9–6.4
X band	Military	7.2–7.7	7.9–8.4
Ku band	Commercial	11.7–12.2	14.0–14 5
K band	Commercial	17.7–21.2	27.5–30.0
Ka band	Military	20.2–21	43.5–45.5

Table 3.2 Frequency Bands for Satellite Communication

- In addition to the frequency listed above, a few bands within the frequency range of 0.8–265 GHz are used in aeronautical, marine, and general mobile systems. Also, some bands within the frequency range of 2.5–275 GHz are used for fixed satellite services (FSS) and broadcasting satellite services (BSS).

Geostationary and Geosynchronous Satellite

- The orbital period of the satellite is expressed in terms of mean solar time. A side real day is defined as the time required for the earth to rotate once on its axis relative to the stars. A sidereal day is measured as 23 h, 56 min, and 4.09 s of mean solar time.
- A satellite with a circular orbital period of one side real day, i.e., approximately 24 h, is called a geosynchronous satellite and has an orbit radius of 42164 km. It orbits in the same direction as the earth rotates, i.e., from west to east.
- A geosynchronous satellite orbit is not properly aligned with the equator; the orbit is inclined. When the angle between the orbit and the equator decreases, the magnitude of oscillation is smaller.
- When the orbit entirely lies over the equator, the satellite remains stationary in relation to the earth's surface. The orbit is known as geostationary orbit, and the satellite is known as a geostationary satellite.

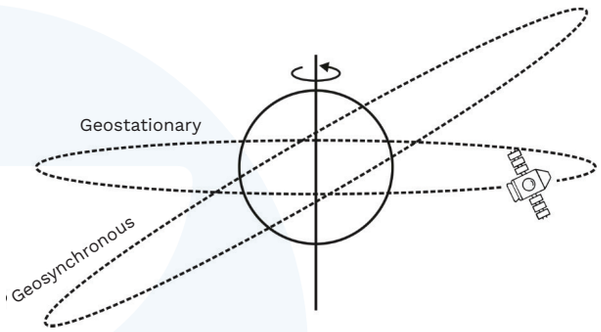


Fig. 2.4

- A geostationary satellite is at an altitude of approximately 35800 km directly above the equator, having a circular orbital period of 24 h. The term geostationary is used as a satellite would appear stationary in the sky as seen by a ground based observer.
- While the geostationary orbit lies on the same plane as the equator, the geosynchronous satellite has a different inclination. This is the key difference between the two types of orbits.
- Geostationary satellites have revolutionised television broadcasting, global tele-communications, and weather forecasting because of their stationary position relative to the earth's rotation.

Global Navigation Satellite System

- GNSS is the standard generic term for a satellite navigation system that provides three dimensional positioning with global coverage. GNSS receivers calculate the position of the object in terms of latitude, longitude,

altitude, speed, direction, and time using a mathematical process called trilateration.

- The established fields for GNSS usage are surveying, shipping, missile guidance, and aviation. The existing GNSSs are
  - The United States 'GPS'
  - Russian 'GLONASS'
  - European 'GALILEO' positioning system
  - China's 'COMPASS' navigation system

GPS Navigation System

- Although compasses and maps have been used to navigate for hundreds of years, GPS gadgets are a recent innovation. For decades, GPS satellites were solely accessible to the military; only in the last few decades did they become available to the general public.
- Many GPS receivers are precise to within 50 ft, which is impressive (15 m). Every day, at both sunrise and sunset, a fleet of 24 satellites circle the planet twice, sending and receiving messages. A GPS receiver uses trilateration to pinpoint its position by locking onto signals from three or more satellites in orbit. By using triangulation, the receiver can pinpoint our precise location.
- In addition, it can calculate our current speed and how long it will take to reach our destination. The receiver determines how much time has passed between when a satellite delivers a signal and when our system receives it, considering data from various signals.
- Many of us have found great convenience in the use of GPS devices. In all likelihood, the growing popularity of GPS devices is music to the ears of many who struggle merely to travel to and from the supermarket. A GPS device can determine not only its exact position but also its velocity and heading. Tracking software for your smartphone's location is available for both Apple's iOS and Google's Android.
- A GPS tracking system consists of a device that employs global positioning satellites (GPS) to pinpoint and record the location of a car, person, or other assets at predetermined intervals. When a position is captured, it may either be retained locally on the tracking device or sent to an online location database through the inbuilt cellular (GPRS or SMS), radio, or satellite modem. It is the job of the GPS module in a GPS tracker to receive the GPS signal and compute the coordinates, allowing the asset's position to be shown against a map background in real time or during the analysis of the track at a later time using GPS tracking software.
- Large capacity is included in data loggers so that coordinates may be stored, while data pushers also have a GSM/GPRS modem for sending coordinates to a control computer through SMS or GPRS ID packets.
- A global positioning system navigator and a mobile phone use the same battery pack. The phone will periodically transmit a text message through SMS or GPRS with the location data gathered by the GPS receiver. Smartphones with built-in GPS and tracking software effectively transform the phone into a data transmitter.
- Data pusher is the most prevalent sort of GPS tracking equipment. Similar to a GPS beacon, this gadget transmits its location and other data, such as speed or altitude, at predetermined intervals to a server that can store and analyse the data in real time.
- To monitor a fleet of commercial vehicles, a GPS tracking system typically consists of two main components: location hardware and tracking software. This set-up is often known as an AVLS or automatic vehicle locating system. Most often, a tracking device is a piece of hardware that is permanently mounted within the vehicle and wired to the car's CAN-bus, ignition system switch, and battery. Data may be collected in more detail, then sent to the GPS monitoring server, and made accessible online through a website where fleet activities can be examined in real time or historically using digital maps and reports.
- Most modern GPS trackers have 'data push' technology, allowing for sophisticated GPS monitoring even in commercial settings. Banks, home delivery services, travel agencies, even schools with a commercial fleet, school buses, and substantial public or private transport networks, including railroads, aeroplanes, and shipping sectors today, utilise GPS.

IRNSS System

- The Indian Regional Navigation Satellite System (IRNSS-1G), now named NAVIC (NAVigation with Indian Constellation), was launched on April 28, 2016, from the Satish Dhawan Space Centre in Sriharikota, Andhra Pradesh, on the Polar Satellite Launch Vehicle (PSLV) C-33.
- Launching the IRNSS-1G satellite means India is now among a small club of nations having its own GPS (Global Positioning System) or navigation system.
- The seventh of the seven satellites that make up the IRNSS space segment, IRNSS-1G is a navigation satellite. Before it, on PSLV-C22, PSLV-C24, PSLV-C26, PSLV-C27, PSLV-C31, and PSLV-C32 in July 2013, April 2014, October 2014, March 2015, January 2016, and March 2016, respectively, were launched IRNSS-1A, 1B, 1C, 1D, 1E, and 1F. With a launch mass of 1425 kg, IRNSS-1G is comparable to other satellites. The configuration of IRNSS-1G is identical to those of its predecessors, IRNSS-1A, 1B, 1C, 1D, E, and F.
- Two navigation and range payloads are aboard the 12 years lifespan satellite.
- With its navigation payload, it will provide customers with signals for navigation services. This payload will use L5 and S band frequencies: the satellite's navigation payload includes an extremely precise Rubidium atomic clock.
- To precisely determine the satellite's distance, the satellite's 'ranging payload' uses a C band transponder.
- In addition to mapping and tracking services, IRNSS's seven satellites give real time data on the location of items to facilitate road, aviation, and marine transportation.
- Positioning data for India and an area out to 1500 km from the coast will be provided by this satellite navigation system.
- Standard positioning services (SPS) will be offered to everyone, while restricted services (RS) will be made available to approved customers only.
- Three of the seven satellites in the system are permanently in orbit around the Earth, while the other four are not.
- The Government of India is looking to the IRNSS as a foundation for introducing its own Global Navigational Satellite Services network, dubbed global indian navigation satellite (GINS).
- The launch of this seventh satellite, which effectively rounds out the originally intended constellation, gives Indian scientists cause for pride in a field where several other nations have more than 20 spacecraft dedicated to navigational purposes. With the use of this constellation, ISRO will be able to extend its services beyond those just intended to benefit the general populace.

Practice Exercise

1. In the spread spectrum technique, multiple users, are assigned:
(A) Same spectrum and same PN code.
(B) Same spectrum and different PN code.
(C) Different spectrum and different PN code.
(D) Different spectrum and same PN code.

2. LTE standard is developed by:
(A) ITU
(B) IEEE
(C) 3GPP
(D) IETF

3. LTE supports:
(A) Frequency division duplexing (FDD)
(B) Time division duplexing (TDD)
(C) Both A and B
(D) None of the above

4. LTE goals are:
(A) Improving spectral efficiency
(B) Lowering the cost
(C) Improving services
(D) All of the above

5. Which of the following is/are functions of IRNSS?
I. It will be useful for navigation.
II. It will be India's very own GPS.
III. It will help in telecommunication.
IV. It will be useful for educational purposes.
(A) I and II
(B) I and III
(C) I, III, and IV
(D) All of the above

6. GPS is based on a principle called:
(A) Transmission
(B) Trilateration
(C) Arbitration
(D) Globalisation

7. Which of the following type of computer is mainly associated with the conversion of analogue output into digital forms?
(A) Analog computer
(B) Digital computer
(C) Mainframe computer
(D) Hybrid computer

8. Consider the following statements:
I. Quantum computer makes direct use of superposition and entanglement to perform operations on data.
II. Quantum computers can solve problems much faster than any classical computer.
Which of the given statements is/are correct?
(A) Only I
(B) Only II
(C) Both I and II
(D) Neither I nor II

9. Who is recognised as the 'father of the modern digital computer'?
(A) George Stibitz
(B) Clifford Berry
(C) Nikofay Brusentsov
(D) Claude Ramsay

10. Microcomputer hardware consists of three basic categories of physical equipment. They are:
(A) Keyboard, monitor, and hard drive
(B) System unit, input/output, and memory
(C) System unit, input/output, and secondary memory
(D) System unit and primary and secondary memories

Answer Key

1. (B) **2.** (C) **3.** (C) **4.** (A) **5.** (A) **6.** (B) **7.** (A) **8.** (C) **9.** (A) **10.** (C)